

Software Requirements Specification (SRS)

For the Project:

GUI-Based Aggregated Recon Tool for Web Pentesting

Prepared by:

Divyanshu Saini (23FE10CSE00278)

Department of Computer Science and Engineering

February 4, 2026

Project Title	GUI-Based Aggregated Recon Tool for Web Pentesting
Version	2.0 (Draft)
Confidentiality	Academic Submission (College)
Institution	Manipal University Jaipur

Certificate

This is to certify that the project report titled ”**GUI-Based Aggregated Recon Tool for Web Pentesting**” is a bona fide record of work carried out by **Divyanshu Saini** (Reg. No.: **23FE10CSE00278**) under my supervision in partial fulfillment of the requirements for the course/project.

Guide: Dr. Sakshi Shringi

Date: February 4, 2026

Signature: _____

Acknowledgement

I express my sincere gratitude to my project guide, faculty members, and peers for their guidance and support during the preparation of this SRS. I also thank the maintainers of open-source security tools and frameworks that informed the design and development of this reconnaissance tool. Special thanks to the cybersecurity community for their valuable resources and documentation that contributed to this project.

Contents

Certificate	i
Acknowledgement	ii
Revision History	1
1 Introduction	2
1.1 Purpose	2
1.2 Document Conventions	2
1.3 Intended Audience and Reading Suggestions	2
1.4 Product Scope	2
1.5 References	3
2 Overall Description	3
2.1 Product Perspective	3
2.2 Product Functions	3
2.3 User Classes and Characteristics	3
2.4 Operating Environment	5
2.5 Design and Implementation Constraints	5
2.6 User Documentation	6
2.7 Assumptions and Dependencies	6
3 External Interface Requirements	6
3.1 User Interfaces	6
3.2 Hardware Interfaces	7
3.3 Software Interfaces	8
3.4 Communications Interfaces	8
4 System Features	9
4.1 Target Management System	9
4.1.1 Description and Priority	9
4.1.2 Stimulus/Response Sequences	9
4.1.3 Functional Requirements	9
4.2 Workflow Management System	10
4.2.1 Description and Priority	10
4.2.2 Stimulus/Response Sequences	10
4.2.3 Functional Requirements	10
4.3 Reconnaissance Tool Integration	10
4.3.1 Description and Priority	10
4.3.2 Stimulus/Response Sequences	10
4.3.3 Functional Requirements	11
4.4 Report Generation System	11

4.4.1	Description and Priority	11
4.4.2	Stimulus/Response Sequences	11
4.4.3	Functional Requirements	11
5	Other Nonfunctional Requirements	12
5.1	Performance Requirements	12
5.2	Safety Requirements	12
5.3	Security Requirements	13
5.4	Software Quality Attributes	13
5.5	Business Rules	13
6	Other Requirements	14

Revision History

Name	Date	Reason For Changes	Version
Divyanshu Saini	19 August 2025	Initial draft	1.0
Divyanshu Saini	February 4, 2026	Added ER and DFD 0,1	2.0

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document specifies the requirements for the GUI-Based Aggregated Recon Tool for Web Pentesting, version 1.0. This document covers the complete system functionality including tool integration, workflow management, and report generation capabilities. The tool serves as a unified platform that orchestrates multiple reconnaissance utilities to streamline the information gathering phase of web application penetration testing.

1.2 Document Conventions

This SRS follows IEEE standard conventions for software requirements documentation. All requirement statements are written in active voice using "shall" to indicate mandatory requirements. Priority levels are defined as High (essential for MVP), Medium (enhanced functionality), and Low (advanced features). Requirements are uniquely identified with tags following the format REQ-X.Y where X is the section number and Y is the requirement number.

1.3 Intended Audience and Reading Suggestions

This document is intended for software developers implementing the reconnaissance tool, project managers overseeing development, quality assurance teams responsible for testing, security professionals who will use the tool, and stakeholders evaluating project requirements.

Developers should focus on Sections 3, 4, and 5 for detailed technical specifications. Project managers should emphasize Sections 1, 2, and 6 for scope and planning considerations. End users should concentrate on Sections 2 and 4 to understand system capabilities and features.

1.4 Product Scope

The GUI-Based Aggregated Recon Tool aims to enhance penetration testing efficiency by providing an integrated platform for web application reconnaissance. The tool combines popular security utilities including Nmap, subdomain enumeration tools, directory fuzzers, Nikto, and SQLMap into a single interface with automated workflow execution and professional reporting capabilities.

Benefits include reduced manual effort in reconnaissance activities, consistent methodology application across assessments, and generation of comprehensive reports for technical and executive audiences. The tool supports the cybersecurity community by standardizing reconnaissance processes while maintaining flexibility for custom workflows.

1.5 References

- IEEE Std 830-1998, IEEE Recommended Practice for Software Requirements Specifications
- OWASP Web Security Testing Guide v4.2
- NIST SP 800-115, Technical Guide to Information Security Testing and Assessment
- Nmap Network Discovery and Security Auditing Documentation
- Nikto Web Scanner Documentation
- SQLMap Automatic SQL Injection Tool Documentation

2. Overall Description

2.1 Product Perspective

The GUI-Based Aggregated Recon Tool is a new, self-contained desktop application that serves as a front-end orchestrator for existing open-source security tools. The product operates independently while leveraging established reconnaissance utilities, providing a unified interface for complex multi-tool workflows.

The system architecture consists of a core orchestration engine, tool integration modules, data aggregation components, and reporting subsystems. The tool maintains compatibility with existing security methodologies while introducing workflow automation and result correlation capabilities.

2.2 Product Functions

The major functions that the reconnaissance tool must perform include:

- Target input and validation for IP addresses, domain names, and URL ranges
- Drag-and-drop workflow designer for creating custom reconnaissance sequences
- Integration and execution of network discovery tools (Nmap)
- Subdomain enumeration using multiple techniques and data sources
- Directory and file fuzzing with customizable wordlists and parameters
- Web vulnerability scanning integration (Nikto)
- SQL injection testing capabilities (SQLMap)
- Real-time progress monitoring and execution logging
- Intelligent correlation and deduplication of reconnaissance results
- Automated generation of professional HTML and PDF reports
- Configuration management and profile sharing capabilities

2.3 User Classes and Characteristics

The tool supports multiple user classes with varying technical expertise and usage patterns:

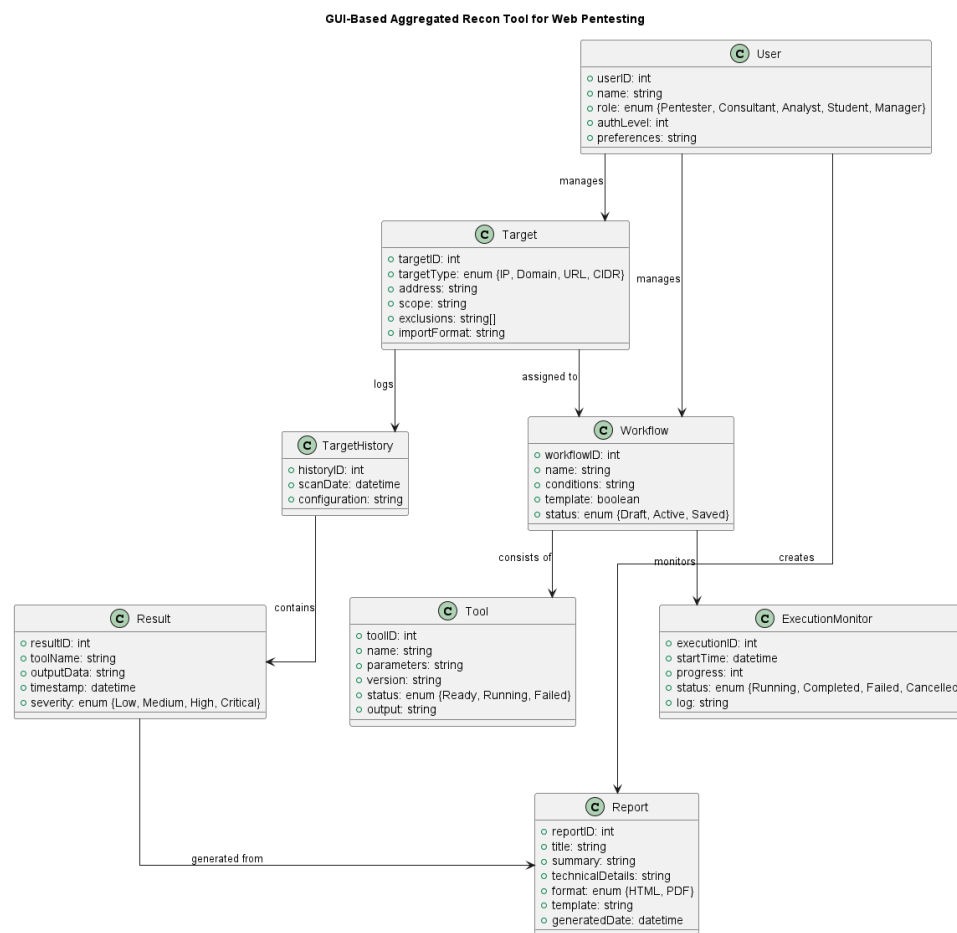


Figure 1: Class Diagram

Professional Penetration Testers (Primary Users): Advanced users with high technical expertise who require comprehensive features, extensive customization options, and detailed reporting capabilities for client engagements. They use the tool regularly and intensively.

Security Consultants (Primary Users): Intermediate to advanced users who need standardized workflows and client-ready reports. They use the tool periodically for security assessments and require ease of use balanced with professional output.

Cybersecurity Analysts (Primary Users): Intermediate users focused on threat assessment and monitoring activities. They require automated workflows, clear result interpretation, and integration with existing security processes.

Security Students (Secondary Users): Beginner to intermediate users learning security methodologies who need guided workflows, educational resources, and simplified interfaces for training scenarios.

IT Security Managers (Secondary Users): Management-focused users who occasionally use the tool for oversight and validation, requiring executive summaries, compliance reporting, and team collaboration features.

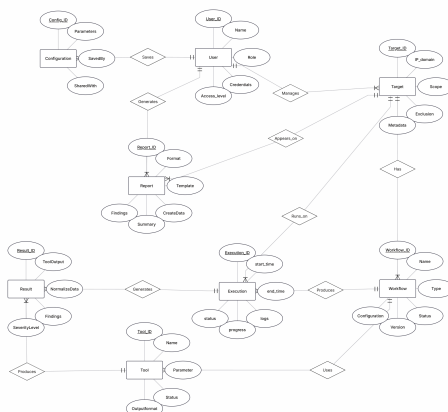


Figure 2: ER Diagram

2.4 Operating Environment

The software will operate in the following environment:

Hardware Platform:

- Minimum: Intel/AMD x64 processor, 4GB RAM, 10GB available storage
- Recommended: Multi-core processor 3GHz+, 8GB RAM, 50GB SSD storage
- Network connectivity for target access and tool updates
- Display resolution minimum 1024x768, recommended 1920x1080

Operating System:

- Windows 10/11 (64-bit)
- Ubuntu 18.04+ or equivalent Linux distributions
- macOS 10.14+ (Mojave or later)

Software Dependencies:

- Python 3.8+ runtime environment
- Java 11+ for certain tool integrations
- SQLite database engine (bundled)
- Native installation of integrated tools (Nmap, Nikto, etc.)

2.5 Design and Implementation Constraints

Several factors will limit design and implementation options:

Regulatory Constraints: The tool must comply with legal frameworks governing penetration testing tools and include appropriate warnings about authorized use only.

Hardware Limitations: Performance is constrained by available system resources, particularly memory usage during large-scale scans and concurrent tool execution.

Technology Constraints: Dependency on third-party tools requires robust error handling, version compatibility management, and platform-specific adaptations.

Security Considerations: All reconnaissance data must be handled securely with encryption at rest and secure credential management for tool configurations.

Usability Requirements: Interface design must accommodate users with varying technical expertise while maintaining professional standards for report generation.

2.6 User Documentation

The following user documentation components will be delivered with the software:

- Quick Start Guide with installation instructions and basic usage
- Comprehensive User Manual covering all features and workflows
- Administrator Guide for enterprise deployments and configuration
- API Documentation for integration with external systems
- Video Tutorials for common use cases and advanced features
- Context-sensitive online help system integrated into the application
- Troubleshooting Guide with common issues and solutions

Documentation will be provided in HTML format for online viewing and PDF format for offline reference. All documentation will be version-controlled and updated with software releases.

2.7 Assumptions and Dependencies

The following assumptions and dependencies could affect the requirements:

Assumptions:

- Users have basic knowledge of penetration testing concepts and methodologies
- Target systems are properly authorized for reconnaissance activities
- Network connectivity is available for accessing targets and downloading updates
- Integrated tools maintain backward compatibility for at least two major versions

Dependencies:

- Availability and continued maintenance of open-source reconnaissance tools
- Python ecosystem stability for core application framework
- Platform-specific package managers for automated tool installation
- Third-party libraries for report generation and data visualization

3. External Interface Requirements

3.1 User Interfaces

The graphical user interface shall provide intuitive access to all system functionality through the following components:

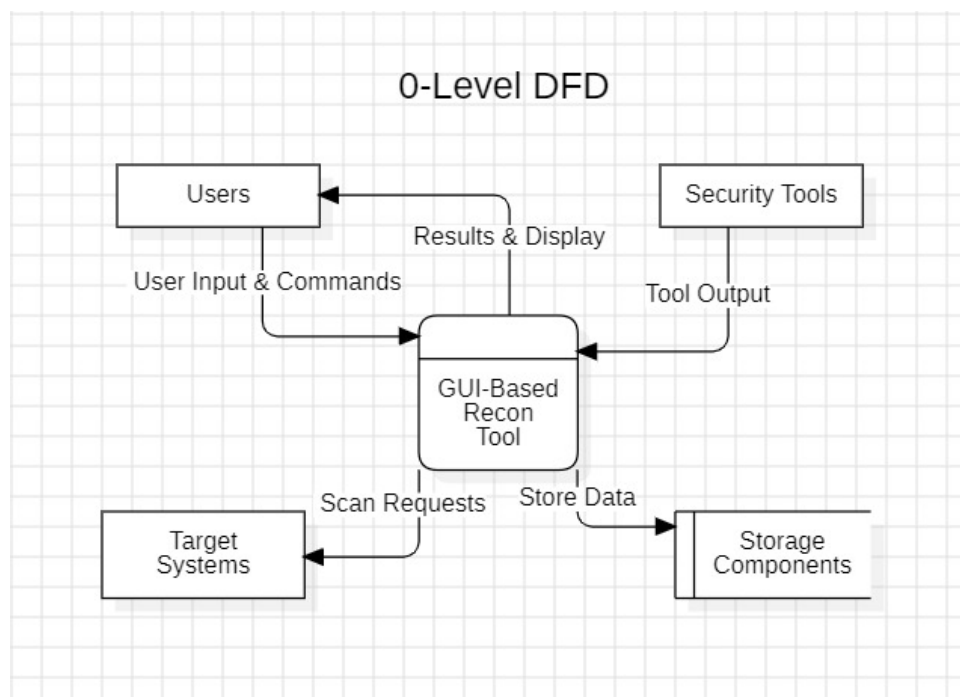


Figure 3: DFD Level 0

Main Dashboard: Central workspace displaying project overview, recent activities, and quick access to common functions. The interface shall follow modern design principles with responsive layouts supporting different screen resolutions.

Target Configuration Panel: Input forms for specifying reconnaissance targets with real-time validation and error messaging. Supports bulk import from files and integration with target databases.

Workflow Designer: Drag-and-drop canvas for creating custom reconnaissance sequences with visual connectors showing tool dependencies and execution flow.

Execution Monitor: Real-time progress tracking interface displaying tool status, completion percentages, and estimated time remaining with detailed logging capabilities.

Results Viewer: Tabular and graphical presentation of reconnaissance findings with sorting, filtering, and search capabilities across all integrated tool outputs.

Report Generator: Template-based interface for customizing report content, formatting, and branding with live preview capabilities.

All interfaces shall include consistent navigation elements, keyboard shortcuts for power users, and accessibility features for users with disabilities.

3.2 Hardware Interfaces

The software interfaces with hardware components through standard operating system APIs:

Network Interfaces: The application utilizes standard TCP/IP stack for target reconnaissance activities, supporting both IPv4 and IPv6 protocols. Network interface selection and configuration are handled transparently through OS networking services.

Storage Interfaces: All data persistence operations use standard file system APIs with support for local and network-attached storage devices. The system requires read/write access to designated data directories.

Display Interfaces: Graphics rendering utilizes hardware acceleration where available through standard display drivers. The interface supports multiple monitor configurations and high-DPI displays.

3.3 Software Interfaces

The system integrates with various software components and external tools:

Operating System Integration:

- Process management for tool execution and monitoring
- File system access for configuration and data storage
- Network socket management for reconnaissance activities
- System service integration for background operations

Database Interfaces:

- SQLite embedded database for local data storage
- Optional PostgreSQL integration for enterprise deployments
- ODBC connectivity for external database integration

Tool Integration APIs:

- Nmap: Command-line interface with XML output parsing
- Nikto: Native API integration with structured result processing
- SQLMap: REST API communication for automated testing
- Custom tools: Plugin architecture supporting standard input/output formats

External Libraries:

- Report generation libraries for HTML and PDF output
- Data visualization components for charts and graphs
- Cryptography libraries for secure data handling

3.4 Communications Interfaces

The application requires various communication capabilities:

Network Protocols:

- HTTP/HTTPS for web-based reconnaissance and API communications
- DNS for domain name resolution and enumeration activities

- TCP/UDP for port scanning and service detection
- SMTP for email-based reporting and notifications

Data Transfer:

- Secure file transfer protocols (SFTP, SCP) for configuration sharing
- RESTful API endpoints for integration with external systems
- WebSocket connections for real-time status updates

Security Requirements:

- TLS 1.3 encryption for all external communications
- Certificate validation for secure connections
- Configurable proxy support for corporate environments

4. System Features

4.1 Target Management System

4.1.1 Description and Priority

High priority feature that manages reconnaissance targets including input validation, scope definition, and historical tracking. Essential for all reconnaissance activities.

4.1.2 Stimulus/Response Sequences

1. User enters target information (IP, domain, URL range)
2. System validates input format and accessibility
3. User defines scope boundaries and exclusions
4. System stores target configuration and displays confirmation
5. User can retrieve and modify target settings for future use

4.1.3 Functional Requirements

REQ-4.1.1: The system shall accept target input in formats including single IP addresses, CIDR notation, domain names, and URL lists.

REQ-4.1.2: The system shall validate all target inputs and provide specific error messages for invalid entries including format errors and unreachable targets.

REQ-4.1.3: The system shall maintain a searchable database of target histories with associated metadata including scan dates, configurations, and results.

REQ-4.1.4: The system shall support exclusion lists to prevent scanning of specified IP ranges, domains, or specific hosts.

REQ-4.1.5: The system shall provide target import functionality from CSV files and popular reconnaissance tool formats.

4.2 Workflow Management System

4.2.1 Description and Priority

High priority feature providing drag-and-drop workflow design capabilities with pre-configured templates and conditional execution logic.

4.2.2 Stimulus/Response Sequences

1. User accesses workflow designer interface
2. User drags tools from palette to design canvas
3. User connects tools to define execution sequence
4. User configures tool parameters and conditions
5. System validates workflow and highlights any issues
6. User saves workflow template for future use

4.2.3 Functional Requirements

REQ-4.2.1: The system shall provide a visual workflow designer with drag-and-drop functionality for arranging reconnaissance tools.

REQ-4.2.2: The system shall include pre-configured workflow templates for common assessment scenarios including web application, network, and API testing.

REQ-4.2.3: The system shall validate workflow configurations and detect circular dependencies or invalid tool combinations.

REQ-4.2.4: The system shall support conditional execution paths based on previous tool results and user-defined criteria.

REQ-4.2.5: The system shall enable workflow templates to be saved, loaded, shared, and version-controlled.

4.3 Reconnaissance Tool Integration

4.3.1 Description and Priority

High priority core feature integrating popular reconnaissance tools including Nmap, sub-domain enumeration, directory fuzzing, Nikto, and SQLMap.

4.3.2 Stimulus/Response Sequences

1. User initiates reconnaissance workflow execution
2. System launches integrated tools according to workflow sequence
3. Tools execute against specified targets with configured parameters
4. System monitors tool progress and captures all output
5. Results are parsed and normalized into common data structures
6. User receives notification of completion and accesses results

4.3.3 Functional Requirements

REQ-4.3.1: The system shall integrate Nmap for comprehensive network discovery and port scanning with support for all scan types and timing options.

REQ-4.3.2: The system shall perform subdomain enumeration using multiple techniques including DNS brute-forcing, certificate transparency logs, and search engine queries.

REQ-4.3.3: The system shall conduct directory and file fuzzing with customizable wordlists, HTTP methods, and response analysis.

REQ-4.3.4: The system shall integrate Nikto for web vulnerability scanning with configurable plugin selection and custom signature updates.

REQ-4.3.5: The system shall support SQLMap integration for automated SQL injection detection and exploitation with safety controls.

REQ-4.3.6: The system shall provide real-time progress monitoring with estimated completion times and the ability to pause or cancel operations.

REQ-4.3.7: The system shall implement comprehensive error handling and recovery mechanisms for tool failures or network interruptions.

4.4 Report Generation System

4.4.1 Description and Priority

Medium priority feature that aggregates reconnaissance results into professional reports with multiple output formats and customizable templates.

4.4.2 Stimulus/Response Sequences

1. User selects completed reconnaissance project for reporting
2. User chooses report template and customizes content sections
3. System aggregates and correlates results from all executed tools
4. User previews report content and makes adjustments
5. System generates final report in requested formats
6. User exports or shares completed reports

4.4.3 Functional Requirements

REQ-4.4.1: The system shall generate comprehensive HTML reports with interactive elements including sortable tables and expandable sections.

REQ-4.4.2: The system shall export reports in PDF format with professional formatting, headers, footers, and organizational branding.

REQ-4.4.3: The system shall create executive summaries highlighting critical findings, risk levels, and recommended actions.

REQ-4.4.4: The system shall include technical appendices with detailed tool outputs, raw data, and methodology explanations.

REQ-4.4.5: The system shall support custom report templates with user-defined sections, formatting, and branding elements.

5. Other Nonfunctional Requirements

5.1 Performance Requirements

The system shall meet the following performance criteria under normal operating conditions:

Response time for user interface interactions shall not exceed 2 seconds for standard operations including navigation, configuration changes, and data retrieval.

Target validation and workflow configuration shall complete within 5 seconds for typical inputs including up to 100 targets and standard tool configurations.

Report generation shall complete within 30 seconds for datasets containing up to 10,000 individual findings with full correlation and analysis processing.

The system shall support concurrent execution of up to 5 different reconnaissance tools without significant performance degradation.

Memory utilization shall not exceed 70% of available system resources during peak operations, with automatic resource management to prevent system instability.

Network bandwidth utilization shall be configurable with rate limiting options to prevent network congestion and detection by monitoring systems.

5.2 Safety Requirements

The system shall implement safeguards to prevent unintended consequences during reconnaissance activities:

All reconnaissance operations shall include confirmation dialogs for potentially intrusive or high-impact scans that could affect target system performance.

The system shall maintain rate limiting and throttling mechanisms to prevent denial-of-service conditions against target systems.

Default tool configurations shall emphasize safety and stealth over speed and comprehensiveness to minimize risk of detection or system impact.

The system shall provide clear warnings about legal and ethical requirements for authorized testing and obtain user acknowledgment before proceeding.

Emergency stop functionality shall be available for immediately halting all reconnaissance activities in case of unexpected issues or unauthorized activity detection.

5.3 Security Requirements

Security measures shall protect both the reconnaissance tool and the data it processes:

All stored reconnaissance data shall be encrypted at rest using AES-256 encryption with secure key management and regular key rotation.

Database connections and network communications shall use TLS 1.3 or equivalent encryption protocols with certificate validation.

User authentication shall support strong password policies, optional multi-factor authentication, and session timeout mechanisms.

The system shall implement role-based access control for multi-user environments with granular permission management and audit logging.

Sensitive configuration data including credentials and API keys shall be stored using platform-specific secure storage mechanisms.

All user actions and system events shall be logged with tamper-evident audit trails for compliance and forensic analysis.

5.4 Software Quality Attributes

The system shall exhibit the following quality characteristics:

Reliability: Mean time between failures shall exceed 100 hours of continuous operation with automatic recovery from transient errors and data integrity protection.

Usability: New users shall be able to complete basic reconnaissance workflows within 30 minutes of initial training with intuitive interfaces and comprehensive help systems.

Maintainability: The system architecture shall support modular updates and tool additions with comprehensive unit test coverage exceeding 80% and automated deployment capabilities.

Portability: The application shall operate consistently across Windows, Linux, and macOS platforms with platform-independent data formats and automated installation packages.

Interoperability: The system shall integrate with common security tools and frameworks through standard APIs and data exchange formats.

Scalability: Performance shall degrade gracefully with increasing target scope and data volume, supporting projects with up to 10,000 targets and 100,000 findings.

5.5 Business Rules

The following operational principles govern system behavior:

Only users with appropriate authorization levels may access sensitive reconnaissance functions and target data based on organizational policies and role assignments.

All reconnaissance activities must be conducted against explicitly authorized targets with documented permission and scope approval.

Assessment data retention shall comply with organizational policies and regulatory requirements with automatic archival and deletion capabilities.

Tool integration shall prioritize open-source solutions with proper licensing compliance and regular security updates.

Report generation shall automatically redact or anonymize sensitive information according to configurable privacy policies.

The system shall maintain compatibility with established penetration testing methodologies and industry best practices.

6. Other Requirements

Database Requirements: The system shall use SQLite for local deployments and support PostgreSQL for enterprise installations. Database schema shall support concurrent access, transaction integrity, and automated backup procedures.

Installation Requirements: Automated installation packages shall be provided for all supported platforms with dependency management, configuration validation, and roll-back capabilities. The installer shall detect and resolve common compatibility issues.

Internationalization Requirements: The user interface shall support localization for major languages with Unicode text handling and culturally appropriate formatting for dates, numbers, and reports.

Legal Requirements: The system shall comply with relevant cybersecurity regulations and export control restrictions. All integrated tools and libraries must have appropriate licenses for commercial and government use.

Accessibility Requirements: The interface shall conform to Web Content Accessibility Guidelines (WCAG) 2.1 Level AA standards with keyboard navigation, screen reader compatibility, and adjustable visual elements.

Training Requirements: Comprehensive training materials shall be provided including video tutorials, hands-on exercises, and certification programs for different user skill levels.